

# Electric Go Kart

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## Main goal

Our primary goal was to bring the kart back to fully operational stock condition, which consisted of fixing the electrical system and repairing the broken mechanical components. Our project fully met our primary goal and was able to bring the kart back to a stock setup. Additionally, we were able to perform a motor/speed upgrade and install custom headlights for the kart, which were some additional goals we were setting out to achieve in the beginning of the semester.

## Electrical components

Our first step in restoring the kart was to replace the missing batteries. Since the kart came with 24V motors and controllers, we decided to get 24V replacement batteries. Our battery choice was two 12V 20Ah LiFePO4 batteries due to their cost, size, and LiFePO4 batteries having excellent weight to capacity ratio. We also installed a new 30A automotive fuse in case of overcurrent. This set up with the stock 24V motor was able to get the kart back to working condition. Our upgrade to the kart consisted of a 36V 1000W motor and a 1500W voltage boost module (later replaced with an 1800W voltage boost module). Headlights were also added to the front of the kart with the wiring running underneath. This setup at 36V boost increased the top speed by around 15%, while increasing the boost to 48V nearly doubled our top speed from the 24V stock setup. A switch was also added to the back of the kart with a custom mount to serve as an e-stop.

## Mechanical components

The mechanical components we designed and manufactured for the kart were as follows: axle keys, wheel spacers, battery mount, motor controller mount, power switch mount, main drive sprocket & chain, headlight covers & mounts, and light switch mount. We CADed the part

mounts and headlight housing to the specific dimensions of the kart and 3D printed them out of PLA. Along the axle, we reengaged the braking system by manufacturing steel keys for the brake rotor and wheels. We also manufactured and installed aluminum spacers to fix the alignment of the wheels along the axle. We replaced the 11-tooth sprocket on the motor with a 13-tooth one to decrease the gear ratio and optimize the kart's speed; we then measured and installed a new chain for the 13-tooth sprocket. Other mechanical work we performed on the kart included angle grinding parts of the frame to accommodate the larger batteries and motor, as well as drilling screw holes in new locations in order to properly mount these parts.

## Troubleshooting

### Electrical

Our test with the initial 1500W voltage boost module was successful, until the main switching MOSFET blew after being stressed by continuous over current from the extremely low gear ratio we were testing at the time. The replacement 1800W voltage boost module comes equipped with a safety feature that cuts power to the motor when continuous over current is detected.

### Mechanical

We adjusted the gear ratio and configuration in order to optimize the kart's top speed and acceleration. Given that our motor runs at 4000 RPM at 48V and our tire circumference is 628.318 millimeters, we calculated and tested the kart's top speed in each of three situations:

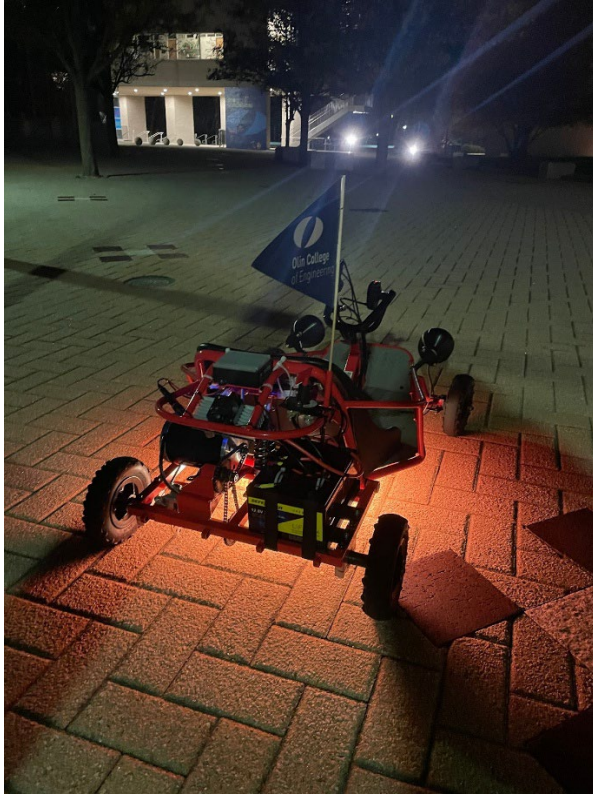
- a. 11T input to 32-to-11 intermediary to 32T output (original configuration)
  - b. 11T input to 32T output (direct configuration)
  - c. 13T input to 32-to-11 intermediary to 32T output (optimal configuration)
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- a.  $\text{Speed} = 4000\text{rpm} / (32/11 * 32/11) * 628.318\text{mm} = 296,978.4296875\text{mm}/\text{min} = \mathbf{11.07\text{mph}}$
  - b.  $\text{Speed} = 4000\text{rpm} / (32/11) * 628.318\text{mm} = 863,937.25\text{mm}/\text{min} = \mathbf{32.2\text{mph}}$
  - c.  $\text{Speed} = 4000\text{rpm} / (32/13 * 32/11) * 628.318\text{mm} = 351,000\text{mm}/\text{min} = \mathbf{13.1\text{mph}}$

While our calculated speeds are not exact because they don't account for a load nor the actual capabilities of our motor, they demonstrate the comparison between various gear ratios and configurations. Because the original configuration (a) did not have a high top speed, we changed to the direct configuration (b). We reached a maximum top speed in this setup; however, we found acceleration to be very poor because the direct configuration required more torque than the motor could provide. Consequently, running the motor for an extended amount of time in this configuration stressed the boost converter with continuous over current and caused it to fail. Therefore, we reverted back to a configuration with an intermediary sprocket, but replaced the 11-tooth main drive sprocket with a 13-tooth one (c). We found this configuration to be the optimal setup to increase top speed while still allowing for adequate acceleration and causing lower stress on the electrical components.

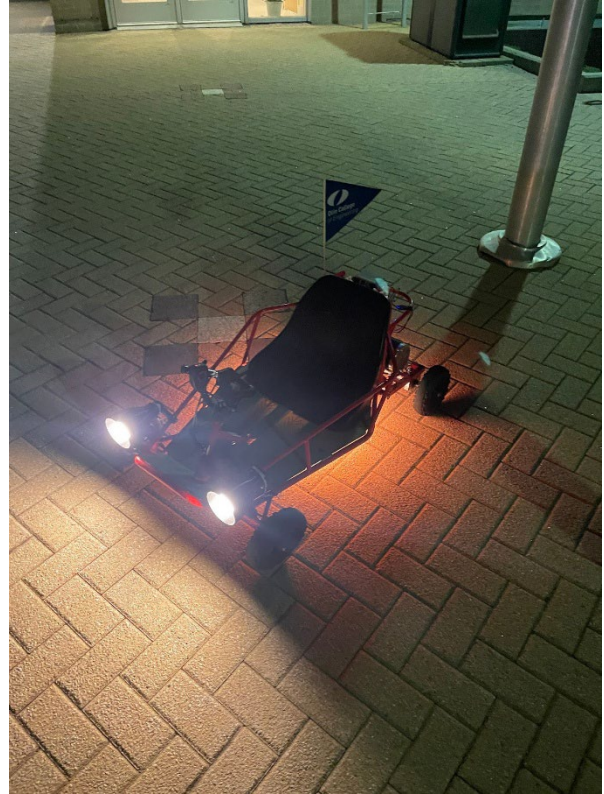
## Photos



Professor Erhardt, our project advisor, riding the go kart.



(Top left) Rear image of the kart



(Top right) Overhead image of the kart

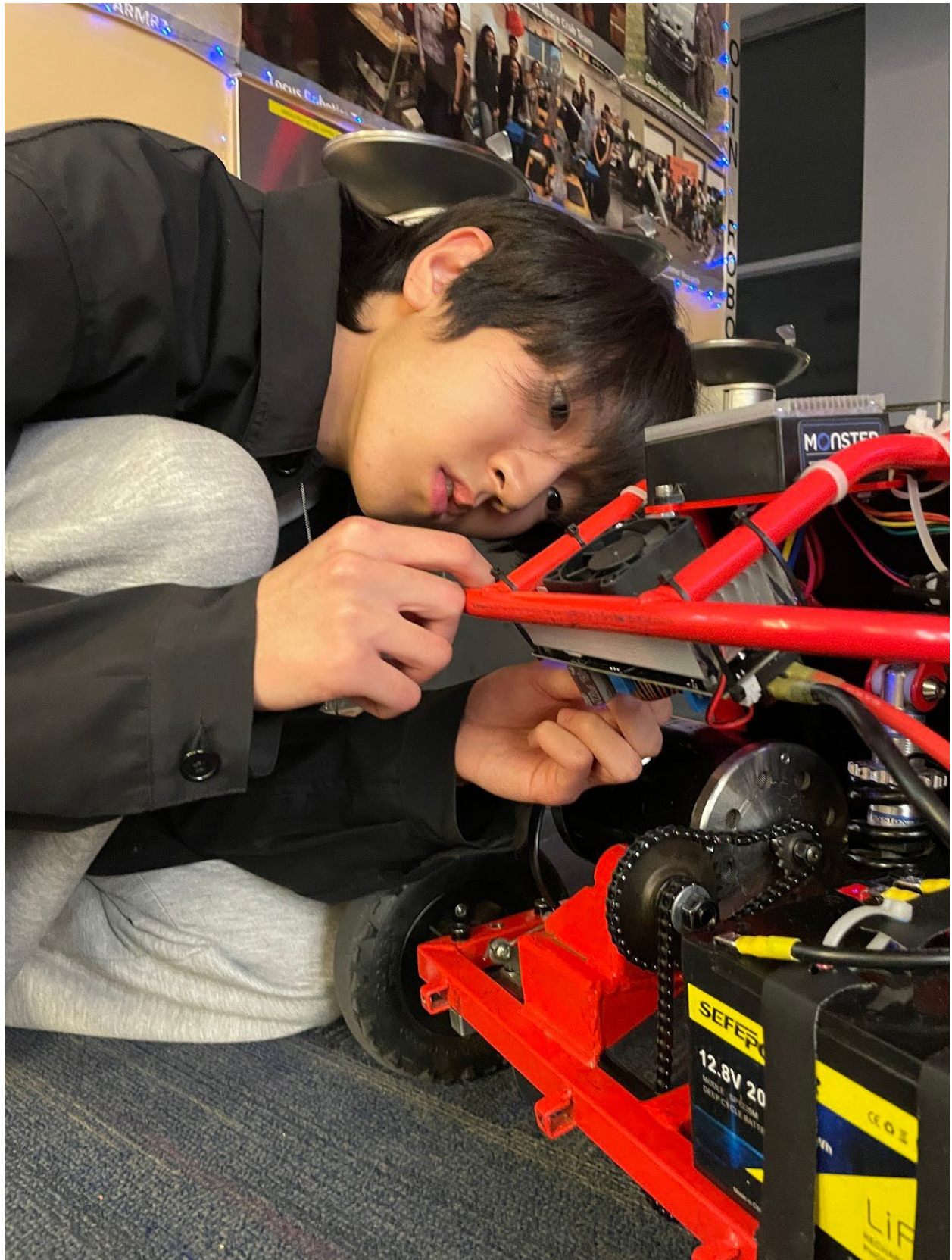


(Bottom left) Testing the kart's weight limit



(Bottom right) Mario kart

Adjusting the boost converter

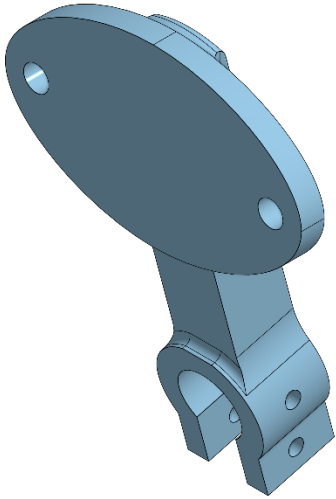


Angle grinding the chassis

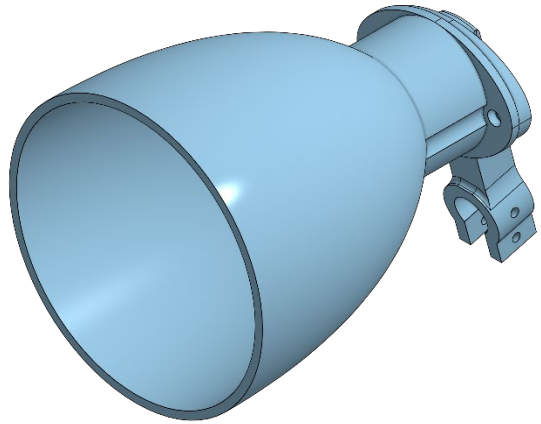


## CAD components

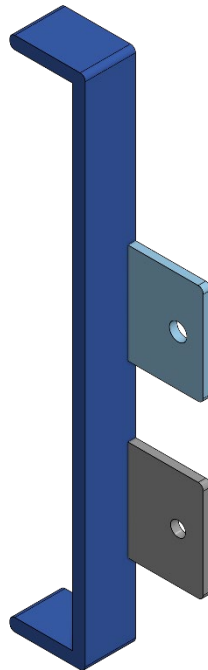
A) Headlight mount



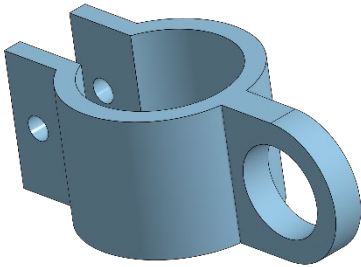
B) Headlight housing assembly



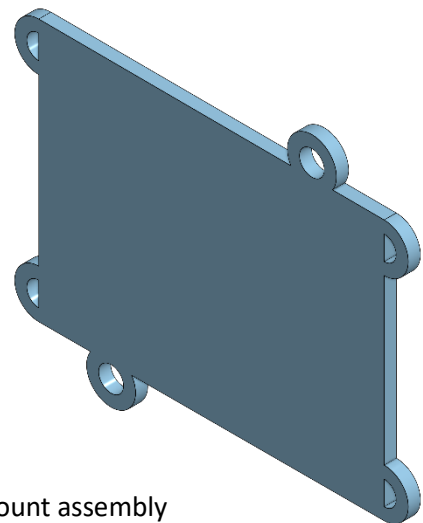
D) Battery mount



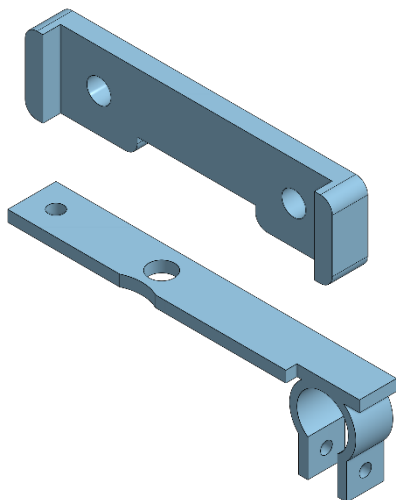
C) Light switch mount



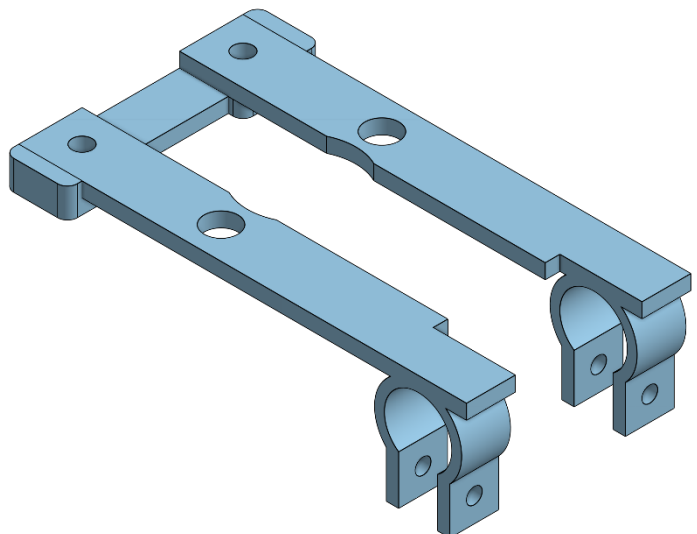
E) Motor controller mount



F) Power switch mount parts



G) Power switch mount assembly



Driving the kart



